Process Transfer Plan

A-Mab Drug Substance — PTP-001

Process Development / Manufacturing Science & Technology (MSAT)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

ADCC, antibody dependent cellular cytotoxicity; AEX, anion exchange chromatography; AMV, analytical method validation; CEX, cation exchange chromatography; CPP, critical process parameter; CQA, critical quality attribute; DoE, design of experiments; DS, drug substance; ELISA, enzyme-linked immunosorbent assay; GMP, good manufacturing practice; GPP, general process parameter; HCP, host cell protein; HILIC, hydrophilic interaction liquid chromatography; HMW, high molecular weight; ICH, International Council for Harmonisation; icIEF, imaged capillary isoelectric focusing; IgG1, a subclass of immunoglobulin G; KPP, key process parameter; LOQ, limit of quantitation; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; NTU, nephelometric turbidity unit; PAR, proven acceptable range; PCMP, Process Characterization Master Plan; PCMR, Process Characterization Master Report; PCP, Process Characterization Plan; PCR, Process Characterization Report; PPQ, process performance qualification; PTP, Process Transfer Plan; qPCR, quantitative polymerase chain reaction; QA, quality assurance; RA, risk assessment; RSD, relative standard deviation; SEC, size exclusion chromatography; SDM, scale-down model; SOP, standard operating procedure;

TCID50, median tissue culture infectious dose; UF/DF, ultrafiltration and diafiltration; WC-CPP, well-controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

1 Purpose and scope

This plan governs the transfer of the A-Mab drug substance manufacturing process from the development site to the commercial manufacturing site. The two sites and the work each one holds are named in §3. The process to be transferred is the fed-batch cell culture and platform purification train described in §2. The transfer covers the manufacturing process, the analytical methods that test the intermediates and the drug substance, and the control strategy that connects the two.

This plan is the parent document of the A-Mab process characterization package. The package has 20 documents and is listed in Table 1.1. Each plan in it commits a study, each report records what that study found, and the set as a whole exists to justify the operating ranges and the control strategy that the receiving site will run. A reader who wants the risk basis for the study scope should start at RA-001. A reader who wants the consolidated outcome should start at PCMR-001.

Table 1.1: The A-Mab document package that this plan parents.

Document ID	Document class	Subject
RA-001	Pre-Characterization Process Risk Assessment	A-Mab Drug Substance
PCMP-001	Process Characterization Master Plan	A-Mab Drug Substance
PCP-003	Process Characterization Plan	Production Bioreactor (Step 3)
PCR-003	Process Characterization Report	Production Bioreactor (Step 3)
PCP-004	Process Characterization Plan	Harvest and Clarification (Step 4)
PCR-004	Process Characterization Report	Harvest and Clarification (Step 4)
PCP-005	Process Characterization Plan	Protein A Chromatography (Step 5)
PCR-005	Process Characterization Report	Protein A Chromatography (Step 5)
PCP-006	Process Characterization Plan	Low-pH Viral Inactivation (Step 6)
PCR-006	Process Characterization Report	Low-pH Viral Inactivation (Step 6)
PCP-007	Process Characterization Plan	Cation Exchange Chromatography (Step 7)
PCR-007	Process Characterization Report	Cation Exchange Chromatography (Step 7)
PCP-008	Process Characterization Plan	Anion Exchange Chromatography (Step 8)

Document ID	Document class	Subject
PCR-008	Process Characterization Report	Anion Exchange Chromatography (Step 8)
PCP-009	Process Characterization Plan	Small-Virus Retentive Filtration (Step 9)
PCR-009	Process Characterization Report	Small-Virus Retentive Filtration (Step 9)
PCP-010	Process Characterization Plan	Ultrafiltration / Diafiltration (Step 10)
PCR-010	Process Characterization Report	Ultrafiltration / Diafiltration (Step 10)
PCMR-001	Process Characterization Master Report	A-Mab Drug Substance

Three activities are outside the scope of this plan. Drug product manufacture, filling and finishing are transferred under a separate plan. Cell bank manufacture and testing are governed by their own records and are not repeated here. The execution of the process performance qualification campaign is also outside this plan, which sets the conditions the campaign must meet before it may start and the strategy it will follow (§7). The campaign itself will be executed under its own protocol and reported separately.

The transfer follows the quality system expectations for technology transfer in ICH Q10 (International Council for Harmonisation 2008) and the lifecycle approach to process validation described by FDA (U.S. Food and Drug Administration 2011). Characterization is the Stage 1 process design work of that lifecycle, and process performance qualification is Stage 2. The enhanced development approach of ICH Q8 (International Council for Harmonisation 2009) and ICH Q11 (International Council for Harmonisation 2012) governs how the design space and the control strategy are defined. ICH Q9 (International Council for Harmonisation 2023b) governs the risk assessment that scopes the studies, and ICH Q5A (International Council for Harmonisation 2023a) governs the evaluation of viral safety. The process model, the quality attribute framework and the risk methodology follow the published case study (CMC Biotech Working Group 2009).

2 Product and process description

A-Mab is a humanized IgG1 monoclonal antibody. Its primary mechanism of action is antibody dependent cellular cytotoxicity, so the fucose and galactose content of the N-linked glycan are linked to efficacy and are treated as quality attributes of the drug substance. The glycan is formed in the production bioreactor and is not modified by the purification train, because none of the platform purification steps separates glycosylation variants under the conditions used here (CMC Biotech Working Group 2009). Control of the glycan attributes therefore sits in cell culture, and the transfer of the cell culture step carries more product quality risk than the transfer of any single purification step.

The drug substance is manufactured at a production bioreactor working volume of 15,000 L and is delivered by the final formulation step at a target concentration of 75 g/L. The train has 8 unit operations, numbered Step 3 through Step 10. Table 2.1 lists them with the role each plays in the control strategy.

Table 2.1: The A-Mab drug substance process train and the principal role of each step.

Step	Unit operation	Principal role
3	Production Bioreactor	Forms the glycan, charge-variant and aggregate CQAs (design-space step)
4	Harvest and Clarification	Primary recovery and clarification; forms no product-quality CQA
5	Protein A Chromatography	Capture; sets leached Protein A; principal HCP and DNA clearance
6	Low-pH Viral Inactivation	Low-pH hold; sets the cumulative XMuLV (enveloped) clearance
7	Cation Exchange Chromatography	Polish; principal aggregate reduction; major HCP/DNA/leached-PA clearance
8	Anion Exchange Chromatography	Flow-through polish; sets the cumulative MVM clearance; clears XMuLV/HCP/DNA
9	Small-Virus Retentive Filtration	Dedicated small-virus removal; principal MVM clearance
10	Ultrafiltration / Diafiltration	Formulation / mass balance; delivers drug substance at target concentration

The purification sequence is the platform sequence used for the sponsor's earlier humanized IgG1 products. Capture on Protein A is followed by a low pH hold, cation exchange, anion exchange, small virus retentive filtration and ultrafiltration with diafiltration. The low pH hold, the anion exchange step and the small virus retentive filter each carry a viral clearance claim, and the three claims are modular in the sense of ICH Q5A (International Council for Harmonisation 2023a). Each step is studied and claimed on its own. The cumulative clearance across the train is stated only in PCMR-001, and no single step report claims it.

Table 2.2 lists the 10 quality attributes that the transfer must preserve, with the acceptance criterion and the criticality assigned to each. Criticality is expressed as a continuum, not as a binary split into critical and non-critical, following the case study (CMC Biotech Working Group 2009) and the risk principles of ICH Q9 (International Council for Harmonisation 2023b). The attributes ranked high and very high drive the study scope. Afucosylation, galactosylation and high mannose are set by the glycosyltransferase activity of the cell, which rises and falls with the nutrient supply and the dissolved gas concentrations the bioreactor holds. Acidic charge variants accumulate through deamidation. The rate rises with pH and with temperature, and the extent reached rises with the time the product spends in the culture at those conditions. Aggregate forms in the culture and is reduced again by cation exchange, which binds the high molecular weight species more strongly than the monomer and leaves them on the column when the pool is cut. Host cell protein, residual DNA and leached Protein A are removed by the chromatography steps, which either bind them while the antibody flows past or elute them away from the product peak. The two viral clearance attributes are cumulative across the train and belong to no single step.

Table 2.2: Drug substance quality attributes, acceptance criteria and assigned criticality.

CQA	Category	Acceptance	Criticality	Tool #1
Afucosylation	glycosylation	2–13 % of glycans	H	60
Galactosylation (total %Gal)	glycosylation	10–40 % (G1+G2 equiv.)	H	48
High Mannose	glycosylation	3–10 % of glycans	VH	80
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60
Acidic charge variants (deamidation)	charge variant	18–40 % (CEX/icIEF)	VL	4
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36
Residual DNA	process impurity	0–0.001 ng/dose	VL	6
Leached Protein A	process impurity	0–5 ppm	L-M	16
Viral clearance — XMuLV (enveloped)	viral safety	16.7–30 log10 (cumulative)	VH	20
Viral clearance — MVM (parvovirus)	viral safety	8.6–20 log10 (cumulative)	VH	20

3 Sending and receiving sites

The sending site is Novacyte Biologics — Cambridge, MA (Development). The receiving site is Novacyte Biologics — Grafton, WI (Commercial DS). Both belong to the same sponsor, and both operate under one pharmaceutical quality system and one change control procedure. That common quality system is the condition ICH Q10 places on a transfer of this kind (International Council for Harmonisation 2008). A change to a set-point, a range or an analytical method during the transfer will therefore be assessed once and applied at both sites.

Development retains the qualified scale-down models, the execution and the statistical analysis of the characterization studies, and the authorship of the characterization plans and reports. It also retains the process description and the justification of the operating ranges until those are approved. The receiving site takes ownership of commercial manufacture. That ownership covers the master batch record, the operation of the equipment train at commercial scale, the engineering and qualification batches, and the routine testing that supports batch release.

The division is a division of work rather than of accountability. Both sites review and approve the process description, and Quality Assurance at both sites approves each handover listed in §10. Where a characterization study produces a range the receiving site cannot hold, the two sites resolve it together before the range is written into a batch record.

4 Site equivalency analysis

Equivalency was assessed for the elements that can carry a process difference between the two sites. Table 4.1 gives the assessment. The comparison is prospective, so each row states what will be compared and how a difference will be managed rather than a measured outcome.

Table 4.1: Site equivalency assessment for the elements that can carry a process difference.

Element	Sending site	Receiving site	How the difference is managed
Equipment train and scale	Qualified scale-down models of every unit operation	Commercial equipment train at full working volume	The models are qualified against the commercial train before any study is run (§5)
Process control capability	Laboratory controllers with manual intervention available	Automated control with recorded sequences and alarms	Control capability is assessed parameter by parameter against the NOR that each parameter is given
Utilities and buffer supply	Buffers prepared per experiment at laboratory scale	Central buffer preparation and hold, water and clean steam from site utilities	Buffer identity, concentration, pH and conductivity are controlled to one specification at both sites, and hold times are qualified at the receiving site
Analytical capability	Methods qualified and in routine use for characterization	Methods to be transferred and requalified before comparability material is tested	Method transfer and requalification under §6.2
Raw materials	Development lots, released against the same specifications	Commercial lots under supplier qualification	Specifications are common, and any vendor change is assessed through the shared change control procedure

Element	Sending site	Receiving site	How the difference is managed
Personnel and training	Study scientists trained on the scale-down models	Operators trained on the master batch record	Training is completed and verified before the engineering batches are run

The largest difference between the sites is scale. The receiving site operates the production bioreactor at 15,000 L, while the characterization studies run at the scale-down volumes qualified under SOP-1001. In cell culture, scale acts through mixing and through gas transfer. Mixing time rises with vessel volume at constant power per unit volume, so a bolus feed takes longer to disperse and the cells near the addition point sit in a higher local concentration for longer. Carbon dioxide stripping falls as the liquid column deepens, because a sparge bubble that rises through more liquid comes closer to equilibrium with the culture before it reaches the surface. Oxygen crosses the same interface in the opposite direction, from the bubble into the culture, so a change in the gas path does not act on the two transfers in the same way. Oxygen supply and carbon dioxide removal are therefore matched separately when a model is qualified.

In the purification steps, scale acts through packing and through pressure. A chromatography column is scaled by holding bed height and linear velocity constant, which preserves the residence time of the mobile phase but not the wall support that stabilizes a narrow bed, so a wide commercial column can lose plate count at a pressure drop the model tolerates. A filter is scaled by area at constant flux, which holds the transmembrane pressure only while the feed carries the same particle load. §5 describes how a model is qualified against these differences.

The second difference sits in process control. Automated control at the receiving site holds a parameter closer to its set-point than manual control does, which narrows the distribution the commercial process will deliver. A narrower distribution does not invalidate a range, because the characterized ranges are wider than the control capability of either site. The reverse case matters more. Where the receiving site cannot hold a parameter inside the NOR that characterization assumes, either the range must be widened by further study or the equipment must be changed.

The remaining elements in Table 4.1 carry a documentation difference and not a process difference. Buffer preparation is the one element in that group where a study is still required, because central preparation introduces a hold before use that laboratory preparation does not have. That study is listed in §8 and belongs to the receiving site.

5 Scale-down model and comparability strategy

The characterization studies do not run at commercial scale. They run on scale-down models, and every conclusion drawn from them reaches the commercial process through the qualification of those models. Qualification is performed under SOP-1001. It follows the principle stated in the lifecycle guidance, that a laboratory model supports a commercial claim only when the adequacy of the model has been verified and justified (U.S. Food and Drug Administration 2011).

A model is qualified in two parts. Scale-independent variables such as pH, temperature and dissolved oxygen are operated at the values proposed for commercial manufacture. Scale dependent variables such as agitation, gas flow, column bed height, linear velocity and filtration flux are set so that the model reproduces the performance of the commercial train rather than its dimensions. The comparison is then made on the outputs the step controls. For a chromatography step those outputs are step yield, pool volume and the impurity levels in the pool. For the production bioreactor they are viable cell concentration, titre and the product quality attributes the step forms.

Table 5.1 gives the characterization scope. It lists, for each unit operation, how many parameters enter the campaign, how the study type splits between multivariate and univariate work, and which plan and report cover the step.

Table 5.1: Characterization scope per unit operation and the documents that cover it.

Step	Unit operation	Parameters	Multivariate	Univariate	Documents
3	Production Bioreactor	9	5	4	PCP-003 / PCR-003
4	Harvest / Clarification	3	0	3	PCP-004 / PCR-004
5	Protein A Chromatography	6	4	2	PCP-005 / PCR-005
6	Low-pH Viral Inactivation	4	3	1	PCP-006 / PCR-006
7	Cation Exchange Chromatography	5	4	1	PCP-007 / PCR-007
8	Anion Exchange Chromatography	5	4	1	PCP-008 / PCR-008
9	Small Virus Retentive Filtration	2	2	0	PCP-009 / PCR-009

Step	Unit operation	Parameters	Multivariate	Univariate	Documents
10	Ultrafiltration / Diafiltration (formulation)	3	0	3	PCP-010 / PCR-010

37 parameters are in scope across the train. 22 of them will be studied in multivariate designs and 15 in univariate studies. 2 of the 8 unit operations carry no multivariate design at all. Harvest and clarification forms no product quality attribute, and ultrafiltration with diafiltration concentrates the pool and exchanges the buffer without forming or clearing one. For these two steps the ranges will rest on univariate work and on platform experience, so no interaction between two of their parameters will be estimated (§8).

Comparability between the two sites will be demonstrated on material, not on documentation. The receiving site will run engineering batches on the commercial train before the qualification campaign starts. Each batch will be sampled at every step and tested by the transferred methods. The drug substance from those batches will be compared against the acceptance criteria in Table 2.2, and each intermediate will be compared against the in-process limit that the corresponding characterization report establishes for its step.

Comparability is judged against those criteria and not against the mean values observed in development. A development mean is a property of the scale-down model and of the small number of batches behind it, and a commercial batch that differs from it is not thereby different in quality. Prior knowledge from the sponsor's earlier humanized IgG1 products supports the expectation that the platform purification steps behave the same way at both scales (CMC Biotech Working Group 2009). However, that expectation is confirmed by the engineering batches rather than assumed from the platform.

6 Transfer of the process, methods and control strategy

The transfer is delivered in three parts. §6.1 covers the manufacturing process, §6.2 the analytical methods and §6.3 the control strategy. Each part names what is handed over, what condition the handover must satisfy and who approves it.

6.1 The manufacturing process

The process description is the source document for the transfer. It states the sequence of unit operations, the materials, the set-point of every parameter and the range each parameter may move within. The set-points and ranges in it are not final when this plan is approved. They will be confirmed, and in some cases narrowed, by the characterization studies, and the confirmed values will be issued as a revision of the process description before the master batch record is drafted.

The receiving site will draft the master batch record from the approved process description. The batch record must carry every parameter the control strategy classifies as critical or well controlled critical, together with its set-point and its NOR, and it must carry the in-process controls and the sampling points that support them. Where a parameter is monitored but not held to a limit, the batch record should record the measured value without an acceptance criterion attached, so that a change in the value is visible without generating a deviation.

The controlled procedures and method validations that support the transfer are listed in Table 6.1. The procedures that describe the operation of an individual unit operation will be reissued in the receiving site's document system before the engineering batches are run. SOP-1001, SOP-1002 and SOP-4001 govern scale-down model qualification, the design and analysis of the characterization studies, and parameter classification. These three remain development procedures, because the work they govern stays at the sending site.

Table 6.1: Controlled procedures and analytical method validations in scope for the transfer.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP

Reference	Title	Type
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2003	Operation of the Production Bioreactor (Fed-Batch)	SOP
SOP-2004	Cell-Culture Media and Feed Preparation	SOP
SOP-2005	Harvest by Continuous Disk-Stack Centrifugation	SOP
SOP-2006	Clarification by Depth and Sterile Filtration	SOP
SOP-2007	Operation of the Protein A Capture Chromatography Step	SOP
SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization	SOP
SOP-2009	Operation of the Low-pH Viral Inactivation Step	SOP
SOP-2010	Operation of the Cation-Exchange Polishing Chromatography Step	SOP
SOP-2011	Operation of the Anion-Exchange Polishing Chromatography Step	SOP
SOP-2012	Operation and Post-Use Integrity Testing of the Small-Virus Retentive Filtration Step	SOP
SOP-2013	Operation of the Ultrafiltration / Diafiltration (Formulation) Step	SOP
SOP-3010	Released N-Glycan Mapping by 2-AB HILIC-UPLC	SOP
SOP-3011	Size-Variant Analysis by SEC-HPLC	SOP
SOP-3012	Host-Cell-Protein Quantitation by ELISA	SOP
SOP-3013	Charge-Variant Analysis by icIEF	SOP
SOP-3014	Residual Host-Cell DNA by qPCR	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3010	N-Glycan Map (2-AB HILIC-UPLC)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation

Reference	Title	Type
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3015	Turbidity by Nephelometry (NTU)	Method validation
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation
AMV-3017	XMuLV Infectivity Titre (TCID50)	Method validation
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	Method validation
AMV-3019	Protein Concentration by A280 (UV)	Method validation

Handover of the manufacturing process is complete when the process description, the master batch record, the bill of materials and the operator training records have been approved at the receiving site. MSAT at the receiving site owns that approval. Quality Assurance at both sites countersigns it.

6.2 The analytical methods

The methods in scope are the release and characterization methods for the drug substance and the in-process methods that support the control of the intermediates. Table 6.2 lists the 8 methods with the performance each one demonstrated at qualification.

Table 6.2: Analytical methods in scope for transfer, with qualified performance.

Reference	Analytical method	Intermediate precision	Limit of quantitation	Accuracy (%)
AMV-3221	Aggregate by SEC-HPLC (verification-qualified)	3.8	0.2 % HMW	99.1
AMV-3016	Leached Protein A by ELISA	6.5	0.25 ppm	97.4
AMV-3017	XMuLV Infectivity Titre (TCID50)	0.3	1.5 log10 TCID50/mL	96
AMV-3019	Protein Concentration by A280 (UV)	1.2	0.5 g/L	99.6
AMV-3010	Released N-Glycan Map (2-AB HILIC-UPLC)	2.5	0.5 % glycan	99
AMV-3015	Turbidity by Nephelometry (NTU)	4	0.5 NTU	98

Reference	Analytical method	Intermediate precision	Limit of quantitation	Accuracy (%)
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	0.32	1.4 log10 TCID50/mL	96
AMV-3012	Host-Cell Protein by ELISA	9.5	1 ng/mg	95

The precision column is the intermediate precision of the method. For the quantitative assays it is a relative standard deviation in per cent. For the two infectivity titre assays it is a standard deviation in log10 units, which is the form the viral clearance calculation uses. The host cell protein ELISA is the least precise of the quantitative methods at 9.5 per cent, and it is the method that supports the host cell protein clearance argument across all three chromatography steps. Method transfer for that assay should therefore be completed and reviewed before the engineering batches are run.

Each method will be transferred under a written protocol and requalified at the receiving site before it is used to test comparability material. A transfer is successful when the receiving site reproduces the qualified intermediate precision of the method and returns accuracy within the qualified range on a common sample set. Where the receiving site cannot meet the qualified precision, the cause should be investigated and corrected before the exercise is repeated. The method may not be used for in-process or release testing until the exercise has passed.

Analytical Development at the sending site owns the transfer protocol and the comparison of the two data sets. Quality Control at the receiving site executes the transfer. Quality Assurance approves the outcome. The person who executes a method transfer may not approve it.

6.3 The control strategy

The control strategy is the set of controls that together assure the quality of the drug substance. It has three layers in this transfer. Parameter classification places every process parameter in a class and attaches a control to it. In-process limits bound what each intermediate may deliver to the next step. Release testing confirms the attributes of the drug substance against Table 2.2.

Classification is performed under SOP-4001 on the evidence from the characterization studies, so the classes are not known when this plan is approved. Each report classifies the parameters of its own step, and PCMR-001 consolidates the classification for the whole train. A parameter whose variability affects a quality attribute is a critical process parameter, and it is a well controlled critical process parameter where the risk of leaving the design space is low. A parameter that affects process consistency without affecting quality is a key process parameter. The remaining parameters are general process parameters and are controlled by the batch record without a criticality claim (CMC Biotech Working Group 2009).

Viral safety rests on three independent steps and not on any one of them. The low pH hold inactivates enveloped virus, the anion exchange step removes both enveloped and small virus in flow-through mode, and the small virus retentive filter removes small virus by size. Inactivation by pH, removal by charge and removal by size act on three different properties of the virus particle, so a particle that survives one step is no more likely to survive the next, and the log reductions are added (International Council for Harmonisation 2023a). The load conditions each claim depends on, such as the pH and the conductivity of the anion exchange load, are controlled by the batch record. The cumulative clearance is stated in PCMR-001 and compared there with the acceptance criteria in Table 2.2.

The control strategy is transferred when PCMR-001 has been approved and the classifications it consolidates have been written into the master batch record. Until that point the receiving site operates against the ranges in the current process description. That basis is appropriate for engineering batches and is not sufficient for qualification batches.

7 Process performance qualification strategy

Process performance qualification is the Stage 2 activity that follows this transfer (U.S. Food and Drug Administration 2011). It demonstrates that the commercial process, operated by the receiving site with the transferred methods and the transferred control strategy, reproducibly delivers drug substance of the required quality. This plan does not execute that demonstration. It states what must be in place before the campaign may start and the strategy the campaign will follow.

5 consecutive batches will be manufactured at commercial scale under the approved master batch record. Every parameter will be held at its set-point, and no deliberate excursion will be run. The campaign will not challenge the edges of the characterized ranges, because those edges are established at small scale and a qualification batch is not a characterization experiment. However, it is recognized that a campaign of this size cannot capture the variability that commercial manufacture will show over its life.

The campaign may start when the following conditions are met.

- The characterization package is complete and PCMR-001 has been approved.
- Every method used for in-process and release testing has been transferred and requalified at the receiving site.
- The commercial equipment train and the utilities that serve it are qualified.
- The master batch record carries the confirmed set-points, the confirmed ranges and the in-process controls.
- The gaps recorded in §8 are closed, or the residual risk of an open gap has been assessed and accepted in writing by Quality Assurance.

Sampling during the campaign will follow the sampling plans in the per step characterization plans, so that each intermediate is tested against the same in-process limit that characterization established for it. The acceptance criteria for the campaign will be stated in the qualification protocol and are not restated here. PCMP-001 plans the characterization campaign that produces the evidence, and PCMR-001 consolidates it. Both are approved before the campaign starts.

Continued process verification begins when the qualification campaign is complete. The monitoring plan for it will be drafted by the receiving site during the campaign. It may not be finalized until the campaign data are available, because the control limits it applies are derived from the performance the commercial process actually shows (U.S. Food and Drug Administration 2011).

8 Gap analysis

Gaps were identified by comparing what the receiving site holds today against what the qualification campaign requires. Table 8.1 lists each gap with its impact, the action that closes it and the function that owns the action. A gap must be closed before the qualification campaign starts, or its residual risk must be assessed and accepted under the quality system (International Council for Harmonisation 2023b).

Table 8.1: Transfer gaps, their impact, the closing action and the owning function.

Gap	Impact	Action that closes it	Owner
Scale-down models are qualified against the development train only	Characterization conclusions reach commercial scale only through the model	Qualify each model against the receiving site equipment train, and record the comparison in the characterization plan for the step	MSAT, sending site
Analytical methods are not qualified at the receiving site	Comparability results and in-process results cannot be generated or interpreted at the receiving site	Transfer and requalify every method in Table 6.2 under a written protocol	Analytical Development
Harvest and UF/DF carry no multivariate design	Operating ranges for these two steps rest on univariate work and platform experience	State the limitation in PCP-004 and PCP-010, and support the ranges with univariate data and prior knowledge	MSAT, sending site
Parameter classification is not established until PCMR-001 is approved	The master batch record cannot be finalized	Draft the batch record after PCMR-001 is approved, and run engineering batches against the process description	MSAT, receiving site

Gap	Impact	Action that closes it	Owner
Buffer preparation moves from per experiment preparation to central preparation and hold	Buffer hold time and stability are not covered by development data	Qualify buffer hold times at the receiving site before the engineering batches	MSAT, receiving site
Viral clearance claims are established at small scale only	The claims cannot be verified at commercial scale	Retain the modular small scale claims, and control the load conditions they depend on through the batch record	MSAT, sending site
Operators at the receiving site have not run this process	An execution error during qualification would invalidate a batch	Train operators on the master batch record and run engineering batches before qualification	Manufacturing, receiving site

The first two gaps set the schedule, because each requires laboratory work at a site that has not yet performed it and neither can be closed by a document review. The remaining gaps are closed by sequencing, by training or by a bounded qualification study, and none of them requires a change to the process itself. The gap on harvest and UF/DF will still be open when the campaign ends, because no multivariate design is planned for either step at any point in the programme. It is accepted because neither step forms or clears a quality attribute, so an unestimated interaction between two of their parameters cannot move a result in Table 2.2.

9 Roles and responsibilities

The transfer is run by a joint team drawn from both sites. MSAT leads the team at each site, and Quality Assurance approves each handover. Table 9.1 gives the responsibilities this plan assigns.

Table 9.1: Functions, sites and the responsibility this plan assigns to each.

Function	Site	Responsibility
Process Development / MSAT	Sending	Owns the process description, the scale-down models, the characterization plans and reports, and the justification of the ranges
MSAT	Receiving	Owns the master batch record, the equipment and utility qualification, the engineering batches and the qualification campaign
Analytical Development	Sending	Owns the method transfer protocols and the comparison of the sending and receiving data sets
Quality Control	Receiving	Executes the transferred methods and performs in-process and release testing
Statistics	Sending	Owns the design and the statistical analysis of the characterization studies
Manufacturing	Receiving	Executes the engineering and qualification batches under the master batch record
Quality Assurance	Both	Approves the plans, the reports, each handover and the disposition of any open gap

A named individual is assigned to each function in the transfer team roster, which is maintained outside this plan and revised as the team changes. Depending on the size of the receiving site team, one person may hold more than one function. The separations recorded in §6.2 and §11 are the exception and are not waived.

10 Deliverables and schedule

Table 10.1 lists what this plan commits to and the order in which the deliverables arrive. The order is a dependency order rather than a calendar. Dates are held in the transfer project schedule, which is revised as the work proceeds and is not reproduced here.

Table 10.1: Deliverables of the transfer and the dependency order in which they arrive.

Deliverable	Owner	Depends on
RA-001, the pre-characterization risk assessment	MSAT, sending	This plan
PCMP-001, the characterization master plan	MSAT, sending	RA-001
One characterization plan per unit operation, PCP-003 to PCP-010	MSAT, sending	PCMP-001
Scale-down model qualification records	MSAT, sending	The characterization plan for the step
Method transfer protocols and reports	Analytical Development	This plan
One characterization report per unit operation, PCR-003 to PCR-010	MSAT, sending	Execution of the corresponding plan
PCMR-001, the characterization master report	MSAT, sending	Every characterization report
Revised process description with confirmed ranges	MSAT, sending	PCMR-001
Master batch record	MSAT, receiving	The revised process description
Equipment and utility qualification records	MSAT, receiving	This plan
Engineering batch reports	MSAT, receiving	Master batch record and method transfer
Qualification protocol	MSAT, receiving	PCMR-001 and the engineering batch reports

The characterization work sets the duration of the transfer, because the qualification campaign cannot start until PCMR-001 is approved and PCMR-001 cannot be written until every characterization report is complete. Equipment qualification, utility qualification, method transfer and operator training do not wait on that chain. Each of them can start when this plan is approved, and Table 10.1 gives this plan as the dependency for all four.

11 Approvals

This plan takes effect when the functions listed in Table 1 have approved it. A revision is required where the scope of the transfer changes, where a unit operation is added to the train or removed from it, or where a gap in §8 is closed by a route other than the action recorded against it. The author of a revision may not approve it.

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